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Analysis of Requirements and Architecture of the Next Generation Flight Management System

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***Abstract*—The increase in air traffic flow and the increase in environmental awareness have put forward higher requirements for today's air navigation systems. How to apply new technologies to safely solve these problems is a problem that the new generation of air navigation systems needs to address. As the air node in the navigation system, the flight management system plays an important role in ensuring flight safety and reducing environmental impact. Through the analysis of the research status of the air navigation system, the functional requirements of the new generation flight management system are sorted out and analyzed, and the typical functions are further analyzed. Take a flight management system architecture as an example to demonstrate how to meet the functional requirements of the new generation flight management system, and analyze the important modules in the system. On this basis, it is recommended to further study the technical direction of the new generation flight management system.**

Keywords—air navigation system; flight management system; function requirements; architecture

I. Introduction

Global air transportation continues to grow rapidly. In the next 20 years, civil aviation passenger traffic will increase at a rate of 4.6% per year, cargo volume will grow at a rate of 4.2% per year, and air traffic flow will reach 2-3 times the current level [1]. Safety and efficiency have become the theme of civil aviation transportation.

In the context of increasing air traffic flow and increasing attention to environmental protection concepts, and in order to improve the safety level of the global air traffic system and fully tap the existing airspace resources, the International Civil Aviation Organization (ICAO) has established a new performance-based navigation system CNS Based on the concept of /ATM, it is proposed to establish a CNS/ATM and avionics coordination (CNS+A) air navigation environment technical route[2][3]. How avionics equipment, especially flight management systems and modern air traffic management systems work together, is the research focus of the new generation of avionics systems.

The oil crisis in the 1970s led to the emergence of a fuel-saving performance management system that can determine the optimal flight speed and altitude of the aircraft. Later it was integrated with the area navigation system and cross-linked with autopilot and autothrottle. It forms a modern flight management system. The flight management system is the highest form of cockpit automation system, which is equivalent to the air node of the air navigation system.

II. Analysis of Traditional Air Navigation System

The International Civil Aviation Organization (ICAO) believes [3]: In the context of rapid growth in air transportation and limited airspace resources, the establishment of a global and automated air transportation network with air-ground coordination is essential to ensure the safety of civil aviation transportation. In 1983, ICAO proposed a new navigation system (FANS). Based on satellites, data links, computer networks and new automation technologies, advanced communications, navigation and surveillance technologies were introduced in the sky, air, and ground, which were later more commonly called communications, Navigation and surveillance/air traffic management (CNS/ATM) system. In order to guide the development and implementation of the new navigation system, ICAO has issued five versions of the Global Air Navigation Plan (Doc9750), and for the first time in the fourth edition of Doc9750 published in 2013, the adoption of "Aviation System Block Upgrade (ASBU))" as a systematic engineering method for realizing global aviation policy, and proposed a technical route for establishing a CNS/ATM and avionics collaborative environment (CNS+A), and identified four overall performance improvement areas [2] [3]:

1. Airport operations;
2. Globally interoperable systems and data;
3. Optimal capacity and flexible flight;
4. Efficient flight track.

Under the guidance of ICAO's new navigation system program, countries are racing to develop and deploy their own new navigation systems [4]. In 2003, the United States set the goal of a modernized national airspace system for the first time, and proposed the next generation air traffic system (NextGen) [5], jointly participated by the Federal Aviation Administration (FAA), the Department of Transportation, the National Aeronautics and Space Administration (NASA) and other departments. In 2004, the single European air traffic management research project SESAR[5], others include Australia's OneSKY, Japan's CARATS, Brazil's SIRIUS, India's FIANS, etc. [6], as well as China's civil aviation modernization strategy [7]. The idea of these air navigation systems is based on four-dimensional (4D) track operation management and distributed air-ground functions [8], which enhances the overall predictability of the system and is expected to achieve higher efficiency. Each flight segment introduces a new operation mode [9].

In order to support the change of operating mode as shown in Figure 1, the current CNS+A technology needs to

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be updated in design concepts and architecture. The key CNS+A concepts are [6]:

- Operation based on four-dimensional trajectory/intent (TBO/IBO);
- Use performance-based communication (PBC), navigation (PBN), (PBS) to support performance-based operations (PBO);
- Enhanced land-based and satellite-based aeronautical communication systems, with a large number of applications for data link communication;
- Global satellite navigation methods using ground-based enhancement, airborne enhancement and satellite-based enhancement;
- Enhanced land-based and satellite-based surveillance, including automatic correlation monitoring and autonomous separation;
- Wide Area Information Management (SWIM);
- Improve the human-machine interface, obtain better airborne and ground equipment interoperability, and improved automation level;
- Collaborative decision-making (CDM), using more accurate information from the airborne system, allowing the participation of all flight planning and management stakeholders to improve system performance;
- Air Traffic Flow Management (ATFM) and Dynamic Airspace Management (DAM).

III. Analysis Of The Functional Requirements Of The New Generation Flight Control System

A. *Traditional FMS function analysis*

The flight management system (FMS) is the "top-level architecture" [10] of the other airborne subsystems in the civil aircraft avionics system, and is the interface layer that connects the pilot and the underlying avionics. FMS can assist the crew to complete various tasks from take-off to landing, and realize the full voyage of automatic management, monitoring and control of the aircraft. At present, FMS represents the highest level of automation in the avionics field.

Based on the integrated modular avionics (IMA) architecture of the future development trend, FMS includes multi-function display, multi-function keyboard and public computing resources for flight management in hardware. The following functions are performed by software embedded in public computing resources:

1. Navigation management and integrated navigation calculation;
2. Calculation of horizontal and vertical guidance commands;
3. Flight plan generation and optimization algorithm;
4. Performance management and calculation;
5. Dual system management;
6. Management and selection of databases;
7. Built-in testing and working status monitoring;
8. Interface management: instrument display, human-computer interaction and air-to-ground data transmission.

Among them, the horizontal flight plan includes the initialization and horizontal revision of the horizontal flight plan (FPLN). Vertical FPLN also includes the same work. The navigation functions include:

1. Determination of navigation mode, comprehensive navigation calculation;
2. Manual and automatic radio navigation station receiver selection;
3. Initialization and calibration of the inertial reference system (IRS);
4. Navigation satellite selection and position initialization.

B. *Analysis of new generation FMS functional requirements*

In the new generation of air navigation system (FANS), FMS still maintains its mission: to ensure flight safety, reduce pilot labor intensity and the probability of human error, and at the same time, in accordance with the operating requirements of the new navigation system, according to the specific operating environment, automatically follow the optimal Flight status (speed, altitude, track) to achieve the required flight and communication navigation monitoring performance, analyze the requirements from the following aspects:

1. Functional requirements: In the FANS system, the new generation FMS system is considered to be a key driving force for improving the operational efficiency and safety of the air traffic system, while reducing the impact of aviation on the environment [6], which can meet complex and changeable operations, safety and Generate global optimal trajectory under environmental requirements. Through the data link, the new-generation FMS system can interoperate with the future ground-based four-dimensional trajectory planning, negotiation and verification (4-PNV) system to achieve trajectory/intent-based operation (TBO/IBO) [11]. In order to meet the requirements of seamless integration of communication, navigation, surveillance and air traffic management described in the FANS system, in addition to the features already defined in the ARINC 702 performance specification [12], the new generation of FMS systems require comprehensive performance-based operating functions , Comprehensive performance calculation function. At the same time, in the new generation of FMS system, the database storage capacity supporting these functions will also grow rapidly, and the storage content will be more abundant, and more attention will be paid to the versatility of the FMS system, graphical human-machine interface, and stronger management functions [13]. Some advanced FMS functions have been implemented and applied, such as the continuous descent approach application based on trajectory optimization [14]. The implementation experience in several airports such as Los Angeles International Airport has shown that compared with the traditional step descent approach procedure, Continuous descent approach can reduce fuel consumption and noise.

2. System structure: FMS is an important avionics system. Historically, in order to meet the increasingly important position of avionics systems and the needs of more and more complex functions, avionics system architecture has experienced three stages: discrete architecture, joint architecture, and integrated architecture [15]. Integrated Modular Avionics System (IMA) is an important direction for the development of avionics systems for large passenger aircraft in the future, in large aircraft (such as: Airbus A380, Boeing B787, COMAC C919) and new generation aircraft (A320 neo and B737 MAX) Among them, the integrated technology of avionics systems is regarded as one of the

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three core and iconic technologies (materials, engines and integrated avionics) of aircraft [16]. In the IMA system, the rational allocation of the various functional partitions of the FMS is also a direction that needs to be studied.

3. System model: Optimal track generation requires better model and parameter description, which is conducive to applying optimal control theory to track generation. When the FMS system appeared, computer technology developed rapidly, and the all-digital simulation system based on the principle of parallel processing gradually replaced the previous digital-analog hybrid computer simulation method. With the enhancement of computer processing and storage capabilities, as well as the application of numerical analysis and other technologies, currently in the initial stage of development of FMS systems and other avionics systems, mathematical models are generally used to replace actual systems for experiments or to simulate the system itself and the working environment. Changes are simulated and quantitative analysis methods are used to obtain the required information. With the widespread application of the IMA architecture, the previous forms of multiple hardware components have been replaced by embedded software modules on the same computing platform, and the demand for models that can analyze the architecture in simulation has also increased. At present, the semi-formal methods represented by Architecture Analysis and Design Language (AADL) and Unified Modeling Language (UML) have good ease of use and intuitiveness, and the modeling environment and tools are mature. Based on the architecture requirements, it can be used to establish FMS system model [15].

In order to ensure flight safety, airworthiness management agencies and industry standard organizations have issued many airworthiness standards regarding the functional requirements, architecture, and system models of avionics. According to the classification of operations, technology and safety, some of the airworthiness standards related to FMS Aviation regulations and standards are shown in Table 1.

C. *Typical functional requirement analysis*

In terms of various requirements, multiple detailed functional requirements can be divided according to specific FMS functions. Here the navigation function is chosen as an example to analyze the requirements in a performance-based operating environment. In order to support TBO operation, the improvement of navigation functions mainly involves database management, integrated navigation, flight planning, horizontal and vertical navigation [17]:

1. Database management

(1) Manage the navigation database required to support TBO and time-based operation;

(2) Improve the navigation database to meet the requirements of newly added flight segment types and navigation resolutions, and support RNP operation requirements for air routes, terminal areas, and specific flight stages.

2. Integrated navigation

(1) Support multi-constellation multi-frequency satellite navigation mode;

(2) Determine the optimal navigation mode, perform integrated navigation calculations, calculate and monitor actual navigation performance and flight technical errors;

(3) Monitor abnormal behaviors of the navigation system, such as system failures, radio interference;

(4) The actual navigation performance is monitored and alarmed to form a good man-machine interface.

Table 1 Some airworthiness codes and standards related to flight management systems

subject	No.	Title
operation	FAR 25	Airworthiness standards: transport aircraft
	FAR 121	Operational requirements: domestic, logo and supplementary services
	FAR 25.1523 Terms and appendices D	Smallest Crew
	FAA AC 120-100	Aviation crew fatigue
	EASA AMC/GM of Part-CAT	Instruments, data and equipment
	ICAO Doc 8335	Operational inspection, validation and continuous supervision procedures manual
technology	ARINC 702A	Advanced flight management computer system
	FAR 25.1301 to 25.1337 Terms	Equipment and installation
	RTCA DO-236	Minimum Aviation System Performance Standard: Required navigation performance for area navigation
	RTCA DO-238	Ergonomic guidance for data link systems
	RTCA DO-283	Minimum operating performance standard: required navigation performance for area navigation
	RTCA DO-289	Minimum Aviation System Performance Standard: Aircraft Surveillance Application
	RTCA DO-325	Minimum operational performance standards: automatic flight guidance and control systems and equipment
	RTCA DO-361	Minimum operational performance standard: flight interval management
	SAE ARP-4653	Flight management system overview
	SAE AS-5506	Architecture Analysis and Design Language AADL
	ICAO 9869	Performance-Based Communication and Monitoring (PBCS) Manual
	ICAO Doc 9613	Performance-based navigation PBN manual
safety	FAR 25.1309 terms	Equipment, system and installation
	FAA AC 25.1309-1A	System design and analysis
	SAE ARP-4754A	Certification considerations for highly integrated or complex aircraft systems
	SAE ARP-4761	Guiding principles and methods for the safety assessment process of civil airborne systems and equipment

3. Flight plan development

(1) Flight segment constraints: including flight segment type, arrival time, altitude, speed and other constraints;

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(2) The repeatability of the flight path is required. According to the performance of the aircraft, current flight conditions, and external environment, a suitable flight segment can be selected.

4. Horizontal navigation and vertical navigation

Control constraints on horizontal and vertical paths in flight attitude, speed, and altitude.

IV. New Generation Flight Control System Architecture And Analysis

A. *A new generation of flight control system architecture*

The new navigation system requires the aircraft to meet the requirements of time-based operation, that is, to obtain enhanced CNS performance and four-dimensional trajectory (4DT) functions. Obviously, the traditional FMS is difficult to meet the requirements, and a new functional framework is needed to meet these requirements. In the new CNS+A navigation system, the application of new sensor technology and multi-sensor fusion technology can improve flight performance. From the perspective of air traffic management, in order to realize the trajectory-based/intent-based operation (TBO/IBO) and the demand for civil airspace by drones, air traffic management needs to introduce a new management system, including dynamic airspace management, availability and Development of a new generation of FMS for man-machine collision avoidance.

After considering the above functional requirements, a new generation of flight management system architecture is shown in Figure 2 [18]. In this architecture, trajectory prediction is the focus, and it needs to meet the corresponding constraints of flight, communication, navigation and surveillance (CNS) performance, and collision avoidance, and perform multi-objective four-dimensional trajectory optimization.

B. *Analysis of important modules*

Modern avionics and ground systems can realize the planning and real-time execution of 4DT functions, including multi-objective 4DT optimization, negotiation and verification in the future CNS+A environment [18]. In the system shown in Figure 2, the performance of the system is improved through the implementation of integrity monitoring and enhancement to meet the required communication (RCP), navigation (RNP) and surveillance performance (RSP), so as to achieve uniformity in all flight phases. The purpose of meeting the required total system performance. It can also achieve the purpose of autonomous collision avoidance, self-distance, and reduction of environmental impact. In the new-generation flight control system shown in Figure 2, important modules implemented in software form:

1. Four-dimensional trajectory optimization: This module is mainly for short-term maneuver planning trajectory, and the core is a multi-objective optimization algorithm based on multiple models. Under the constraints of the model and constraint modules, optimization is performed based on a variety of performance standards and cost functions. The optimization goals can be minimum fuel consumption, flight time, operating costs, and reduce noise and exhaust emissions. The basis for optimization includes different databases: traditional databases related to FMS include magnetic deviation, performance database, and navigation database; in addition to these databases, meteorological databases, including digital map databases for population, elevation, and environmental information, are also introduced. At the same time, it also includes cooperative and non-cooperative UAV collision avoidance modules, and interfaces with ground and air trajectory negotiation and verification. Therefore, in addition to tactical intervention and emergency collision avoidance tasks, the four-dimensional trajectory optimization module also involves meteorological and airspace flow information.

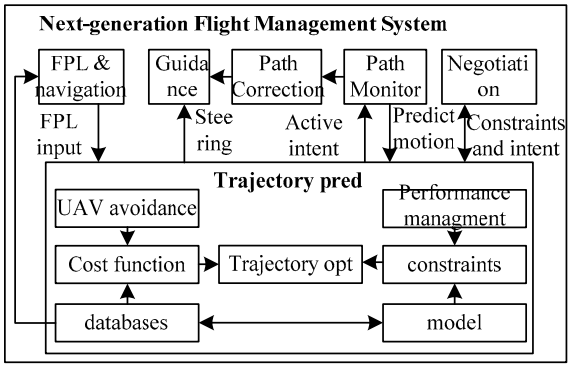

Fig. 1. A new generation of flight management system architecture

2. Performance management: monitor the errors in the 4D trajectory intent, and meet all the performance requirements of communication navigation surveillance (CNS) in the different stages of 4D trajectory negotiation and verification. At the same time, the integrity alert warning sign is generated based on the input from different sensors/systems and the predefined decision logic. For example, determine the main reasons for signal interruption and flight quality degradation of navigation satellites, such as deceptive interference, multipath interference, suppressive interference, antenna occlusion, and poor geometric position of the aircraft relative to the satellite.

3. Four-dimensional trajectory monitoring: Calculate the deviation between the implementation of the four-dimensional trajectory intention and the estimated/predicted aircraft state, and generate horizontal, vertical and time profile deviations.

4. Flight path correction: Correct the path deviation according to the horizontal, vertical and time profile. The generated orientation instructions will be provided to the guidance module.

5. Track negotiation and verification: It executes the negotiation process, which can be initiated by the ground-based four-dimensional trajectory plan negotiation verification system (4-PNV) or the pilot through the FMS man-machine interface. The ground-based 4-PNV system is part of the collaborative decision-making (CDM) network and is connected to the air traffic control department and the airline operations control center (AOC). Through this data communication method, the FMS and the ground have established a negotiation mechanism about the trajectory.

V. Research Direction Analysis

As mentioned above, in the new navigation system, the flight control system, which is the air node with the air navigation system, has added a lot of functional requirements. The following analysis of several key research directions:

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Structure and function division: The trajectory-based operation requires the addition of a new amount of time control, as well as the addition of a new guidance method, and how to make corresponding changes in the structure and interface. The new mode of operation requires real-time weather and traffic control information, and how to ensure smooth communication with air and ground data. The transmission and display of system monitoring information is the design of interfaces with other airborne systems, such as central maintenance, display systems, and alarm systems.

Task and trajectory management: In the new navigation system, taking into account the impact of drones in the airspace and environmental protection, a new task and trajectory management module is needed to achieve safe evasion of maneuver and a more environmentally friendly trajectory. For example, add drone perception and avoidance module; establish a new aircraft performance database to manage climb, cruise and descent profiles, so that the aircraft can operate with lower noise, exhaust emissions, fuel consumption and real-time perception of weather.

System performance monitoring: In the new navigation system, performance is the basis of operation, and performance can be divided into communication, navigation, monitoring and flight performance. Due to the addition of new communication, navigation and monitoring methods and flight control modes, the monitoring of these performances requires the introduction of new databases and performance estimation algorithms. For example, determine the communication navigation monitoring method with the best performance; on the basis of multi-sensor output, establish decision logic, generate integrity, and calculate the actual performance. When the performance drops, it can promptly warn and switch the mode.

Man-machine and communication interface: Compared with traditional flight control systems, the amount of display and communication information of the new generation flight control system will greatly increase. How to effectively display and utilize existing technologies is the focus of the solution. A feasible solution is to use multiple technologies such as tactile feedback, automatic speech recognition and gesture recognition for comprehensive applications to improve practicality and control efficiency. The communication interface can use the current higher-speed interconnected aircraft data gateway to apply it to the flight management system communication under the premise of ensuring the information authentication and safety.

Simulation verification technology: Ground simulation experiment is an important means of current avionics development. It is necessary to select a suitable simulation environment and means, develop operational scenarios that can simulate the new navigation system, and verify the functions and performance of the new generation flight management system. In this process, how to make the verification results meet the airworthiness requirements is also a technical problem that needs to be addressed. (Introduction of take-off and landing based on vision and artificial intelligence)

VI. Conclusions

In the new navigation system, the new generation flight management system as the air node can guide the aircraft to follow the desired trajectory that meets the requirements of time and environmental protection. With the support of stronger environmental sensing and sensing technology, higher-speed digital communications and real-time weather forecasting, aircraft status and flight control are more accurate, and there is more room for airspace management and trajectory negotiation, which increases airspace capacity. This paper analyzes the functional requirements and architecture of the new generation flight management system in the context of the new navigation system, and refines the directions for further research, hoping to provide help for the research and development of flight management systems.

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